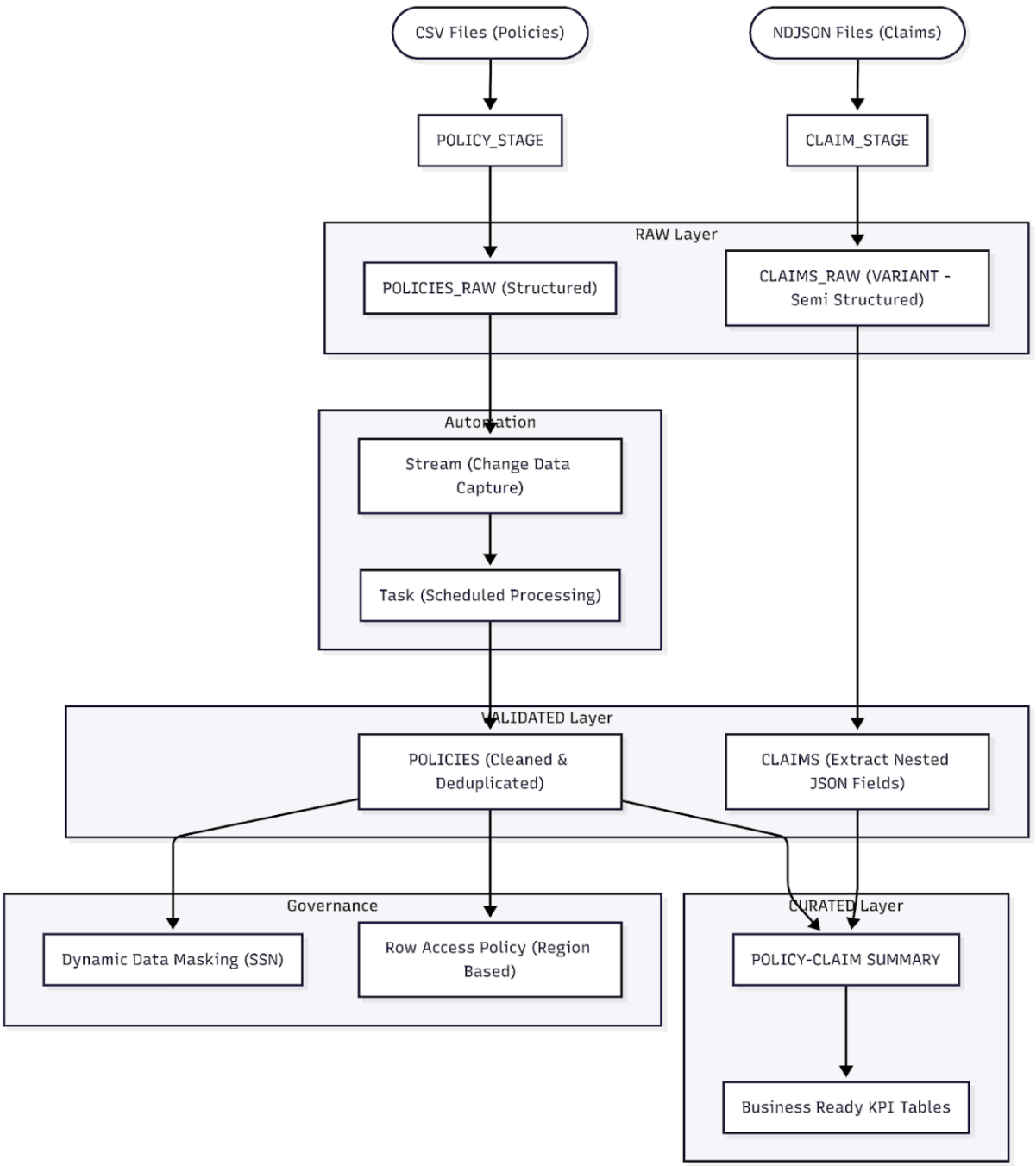
ABC INSURANCE

1. **Project Objective**
   - Scalable Snowflake Data Architecture
   - Robust Data Ingestion & Processing
   - Data Quality & Validation Framework
   - Security & Governance Implementation
   - KPI-Based Reporting & Analytics Enablement

2. **Architecture Overview**
   - Stage Layer – File ingestion.
   - RAW Layer – Stores data as-it-is from the source.
   - VALIDATED Layer – Data cleaning and transformation.
   - CURATED Layer – Business-ready tables for reporting and KPIs.



3. File Handling Strategy

- Policies are received in CSV format (structured data).
- Claims are received in JSON format (semi-structured data).
- Separate file formats are defined for CSV and JSON
- Files are uploaded to internal Snowflake stages before loading

4. **RAW Layer Processing**
    - Policies CSV data is loaded into a structured table.
    - Claims JSON data is loaded into a VARIANT column.
    - RAW layer stores data without transformation.
    - Load timestamps are maintained for audit tracking.

5. **Understanding VARIANT (Semi-Structured Data Handling)**
    - VARIANT is a Snowflake data type used to store semi-structured data like JSON.
    - It enables flexible schema handling (schema-on-read).
    - Nested JSON fields can be extracted using dot notation.
    - Supports arrays and nested objects.

6. **VALIDATED Layer Processing**
    - Remove duplicate records.
    - Apply data quality validations (null checks, date validation, premium validation).
    - Extract structured fields from JSON VARIANT column.
    - Standardize formatting and enforce referential integrity.

7. **Incremental Processing & Idempotency**
    - Incremental files are processed using MERGE logic.
    - Existing records are updated when matched.
    - New records are inserted when not matched.
    - Ensures safe re-runs and prevents duplicate data.

8. **CURATED Layer (Business View)**
    - Join policies and claims data.
    - Create business-ready analytical tables.
    - Prepare datasets for dashboard reporting.
    - Enable KPI calculations.

9. Key KPI Implementations

- Agent Contact Data Quality.
- Premium and Policy Term Validation.
- Policy-Claim Coverage Ratio.
- Average Premium by City.
- Top Cities by Claim Severity.
- Cross-Sell Penetration Rate.
- 30-Day Onboarding Attachment Rate.
- Policy-Only vs Claim-Only Customers.

10. Governance & Security

- Dynamic Data Masking for sensitive fields like SSN.
- Row Access Policies for region-based access control.
- RBAC (Role-Based Access Control) implementation.
- Append-only logs and metadata tracking.

11. Automation Using Streams & Tasks

- Streams capture new or updated records automatically.
- Tasks run scheduled queries to process incremental data.
- This avoids full reload every time.
- It makes the pipeline faster and more efficient.

12. RESULTS

- Handled structured and semi-structured data effectively.
- Implemented idempotent incremental logic.
- Ensured governance using masking and row-level security.
- Built an automated, scalable Snowflake pipeline.